Electrical grid automated visual asset inspection - an application to power line insulators

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Abstract

This paper represents an **initial investigation** to test the efficiency of a **deep learning model** to address the problem of identifying defects on visible light images of **power line insulators**. We tested a publicly available dataset with **1688 images** containing over 6000 shell insulators, using a Faster R-CNN architecture on the Detectron2 framework. We achieved a mean average **recall** (mAR@50:95) of **82,6%** with a **79,9%** mean average **precision** (mAP@50:95).

Motivation

Electric grid asset **inspections are critical** to ensure that modern societies can continue to depend on a stable and uninterrupted supply of electricity. Utility companies are facing an **increasing volume of** collected data such as **images**, during infrastructure inspections, namely **by drones** equipped with sensors.

Its processing is currently being strangulated by human interpretation tasks, mainly because escalating this pipe-line element is expensive and prolonged. However, minor inspection responsibilities can be **automated** using **computer vision** and **deep learning** techniques to support the system's growing **demand**.

Methodology

Our work focus on the central **core** of an automated **inspection** pipeline. The implemented system **receives** high resolution color **images**, obtained in real life scenarios of electric power transmission and distribution, containing different types and amounts of electric insulators, **and generates** a localization **bounding box** with a respective class of the **predicted defect**.

We accessed a publicly available dataset [1] of high-resolution images of transmission line ceramic insulator strings, in a diversified range of colors, captured from various angles and lighting conditions, that contains:

- **1600 images** with 5780 insulator shell assets for **training**
- **88 images** with 300 insulator shell assets for **testing**

Each image is annotated using COCO-style annotations, which include the bounding box top-left coordinates, width, height and a class label designation, **Flashover** damage , **Broken** or **Good** insulator shell. Some examples are presented in Figure 1:

Figure 1. Example of the diversity of insulator image samples

Framework

Our defect detection system relies on a robust computer vision framework developed by Facebook AI Research (FAIR). **Detectron2** [3] stands out for its adaptability and effectiveness across multiple computer vision tasks, including instance segmentation, **object detection**, and keypoint detection. Its modular design simplifies customization for precise specifications and streamlines data handling with utility functions for tasks such as data augmentation, resizing, training, evaluation, and even **architecture selection**.

Architecture

Our architecture selection was a **Faster R-CNN** model [2] that is well-known for addressing the computational bottleneck of its predecessor, Fast R-CNN, replacing a traditional region proposal algorithm for a Region Proposal Network (RPN) that shares the convolutional features for better inference performance.

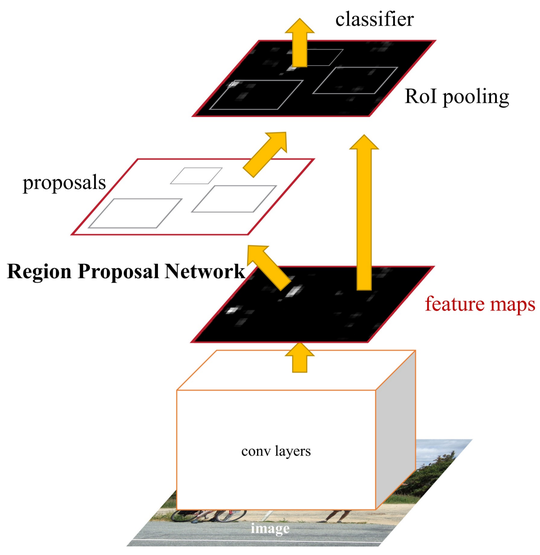

Figure 2. Faster R-CNN and RPN architectural overview [2]

The images feeding the CNN originate features, such as edges, shapes and object parts, that are encoded in feature maps and processed by the RPN, which tests different box shapes through these maps using larger sliding steps. This process results in a set of anchor boxes, each with a respective score related to the probability of an object's existence. Subsequently, the model applies Non-Maximum Suppression (NMS) to eliminate redundant or highly overlapping proposals.

Finally, the classifier block utilizes the proposal regions to efficiently reduce the feature map search. It achieves this through an ROI pooling operation, which converts irregularly sized regions into a fixed-sized format. This approach allows Faster R-CNN to focus its object identification efforts on regions with high probability, rather than inefficiently conducting a sequential or random search.

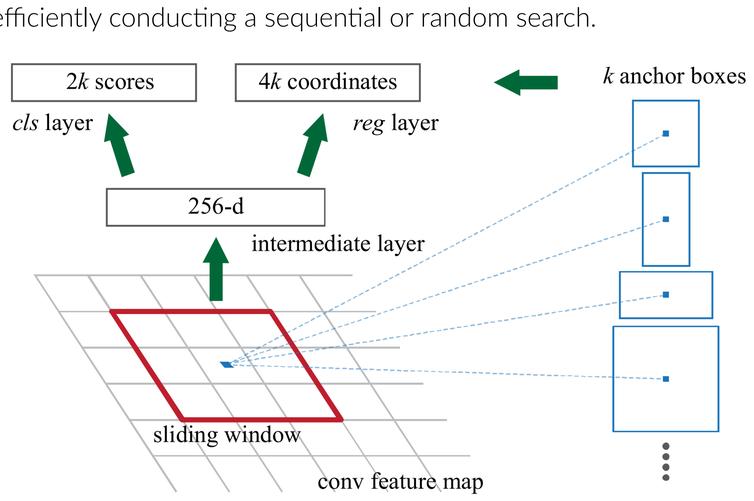

Figure 3. Region Proposed Network (RPN) overview [2]

Results

We conducted **quantitative** evaluations using standard metrics, including the mean Average Precision (mAP) and the mean Average Recall (mAR), across typical Intersection over Union (IoU) thresholds within the MS COCO **benchmark**.

As shown in Table 1, the average **precision** is relatively **high**, and higher for **IoU@50** and IoU@75 threshold limits in all classes. These metrics can not be blindly trusted because of the overlaping objects. This is why we see a noticeable **drop in performance for mAP@50:95**, especially in the 'Broken' class. This drop was expected, as higher threshold values only consider more accurately localized decisions.

Looking at recall, the **'Broken'** class exhibits **the highest recall** in the **initial detection**. This suggests that approximately half of the **top-scored region proposals** are related to broken shell insulators. This result is not surprising, as broken insulators often exhibit more **distinct features** such as sharp white edges.

In the defective classes, within the **first 10 detections** (in the same image), the model identifies a significant portion of all the objects it will eventually detect. An exception to this trend is observed in the **'Good'** or 'No-issues' class, which still experiences a significant **increase in recall** after the 10^{th} detection. This is primarily because some images contain more than 10 good insulator shells.

Table 1. Standard metric scores on test inference

Metric	IoU	max Dets	Class			
			Good	Broken	Flash	All
mAP	@50:5:95	100	0.819	0.726	0.822	0.799
	@50	100	0.966	0.955	0.969	0.951
	@75	100	0.954	0.860	0.965	0.914
mAR	@50:5:95	1	0.108	0.521	0.290	0.305
	@50:5:95	10	0.723	0.781	0.856	0.776
	@50:5:95	100	0.866	0.785	0.870	0.826

Evaluation

Figure 4. Example predictions during test inference

Observing examples of wrong detections, like the first 3 images in Figure 5, we conclude that the model fails to detect a broken shell insulator if it does not have a brighter, sharp edge visible. The second group of 2 images fails to identify good insulators near broken ones, what is not as critical as failing to identify defective ones. Illustrated by the 6^{th} image is the confirmation that allowing a high number of maximum detections can lead to less accurate predictions, with a wrong class overlapping a correct prediction. Next, the misclassification of a good insulator for a broken one suggests that the elliptical shape of a shell insulator does not carry as much weight in the final decision as needed. The model is clearly not well-fitted yet for evening light conditions and prominent shadows due to a lack of training examples (e.g. 8^{th} image). Lastly, translucent glass insulators cannot be correctly detected, due to the absent of training examples.

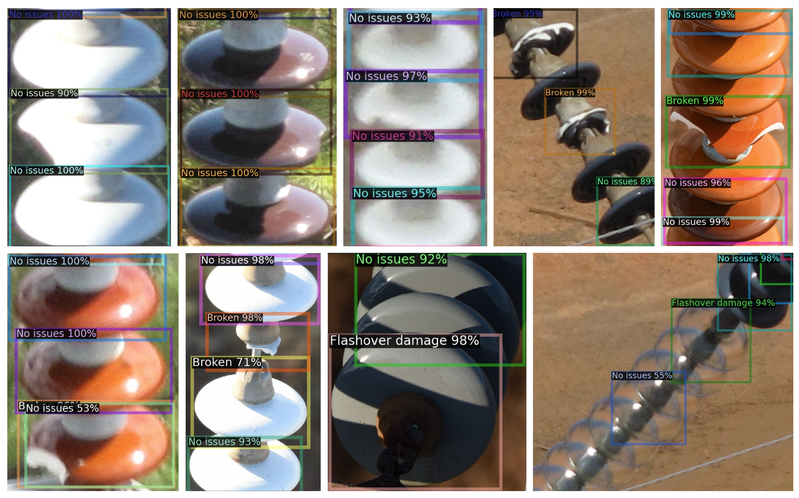

Figure 5. Examples of miss predictions during test inference

Conclusions and future work

The obtained results demonstrate the **high** practical **value** of this approach in the context of automated electrical grid asset inspection. After this successful initial investigation we intend to evolve to a broader project in **collaboration with a Distribution System Operator** (DSO). For the future work, we propose to **adapt the model architecture** in order to enable it to classify different **environmental conditions** such as lighting, focus, and object distance. This approach will allow for precise optimization aiming to enhance performance and robustness under varying conditions. We also aim to integrate inspection results with **maintenance management**.

References

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